

# RITUXIMAB PATIENT KNOWLEDGE ASSISTANT

## Final Project Report & Business Case

*Phase 2 — Azure-Native Production Release*

|                   |                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project:</b>   | Rituximab RAG Clinical Knowledge Assistant                                                                                                              |
| <b>Author:</b>    | Raju Kumar                                                                                                                                              |
| <b>Version:</b>   | Phase 2.0 — Azure Native                                                                                                                                |
| <b>Status:</b>    | <b>Live — Production Deployed</b>                                                                                                                       |
| <b>Azure URL:</b> | <a href="https://rituximab-rag-fmgvg6aravebatbz.centralindia-01.azurewebsites.net">rituximab-rag-fmgvg6aravebatbz.centralindia-01.azurewebsites.net</a> |
| <b>GitHub:</b>    | <a href="https://github.com/raju-AI-portfolio/rituximab-rag-assistant">github.com/raju-AI-portfolio/rituximab-rag-assistant</a>                         |

## 1. EXECUTIVE SUMMARY

The Rituximab Patient Knowledge Assistant is a production-grade, Azure-native Retrieval-Augmented Generation (RAG) application designed to deliver 24/7 source-verified clinical information to patients undergoing Rituximab therapy across oncology, rheumatology, and neurology indications.

The system addresses a critical healthcare delivery gap: over 3 million patients globally receive Rituximab therapy yet lack reliable, always-available access to accurate drug information between clinical appointments. Misinformation, treatment anxiety, and unnecessary emergency contacts result in measurable clinical and economic costs to healthcare providers and payers.

The project has been delivered in two phases. Phase 1 established the core RAG architecture using Pinecone and OpenAI, deployed on Render.com. Phase 2 — completed in May 2026 — migrated the entire stack to Microsoft Azure, replacing all third-party services with enterprise-grade Azure-native equivalents: Azure App Service for hosting, Azure OpenAI GPT-4o for generation, Azure AI Search for vector retrieval, and GitHub Actions for CI/CD automation.

|                                              |                                     |                                     |                                  |
|----------------------------------------------|-------------------------------------|-------------------------------------|----------------------------------|
| <b>1,019</b><br>Medical knowledge<br>vectors | <b>0.80+</b><br>Avg retrieval score | <b>GPT-4o</b><br>Azure OpenAI model | <b>6</b><br>FDA/NIH/NCCN sources |
|----------------------------------------------|-------------------------------------|-------------------------------------|----------------------------------|

## 2. PROBLEM STATEMENT

### 2.1 The Patient Information Gap

Rituximab is one of the most complex and consequential biologic therapies in clinical use. It is prescribed across six major disease categories — Non-Hodgkin Lymphoma, Chronic Lymphocytic Leukemia, Rheumatoid Arthritis, Multiple Sclerosis, Lupus, and ANCA-associated vasculitis — yet patients consistently report inadequate access to reliable drug information.

| Patient Challenge                  | Clinical Impact                             | Economic Cost                                  |
|------------------------------------|---------------------------------------------|------------------------------------------------|
| Cannot reach care team after hours | Delayed reporting of infusion reactions     | Unnecessary ER visits — avg \$2,500/visit      |
| Unreliable internet health sources | Treatment non-adherence from misinformation | Lost therapy efficacy, repeat hospitalisations |
| Complex FDA label language         | Misunderstanding of safety warnings         | Regulatory liability for healthcare providers  |
| Fear of infusion reactions         | Treatment avoidance, missed doses           | Disease progression, re-treatment costs        |
| Vaccine uncertainty on therapy     | Preventable infections (PML, HBV)           | Critical care costs — \$50K–\$200K per event   |

## 2.2 Market Size & Opportunity

The global patient population receiving Rituximab-class biologics represents a significant and growing addressable market for digital health support tools:

| Indication                         | Global Patients    | US Patients      | Annual Growth   |
|------------------------------------|--------------------|------------------|-----------------|
| Non-Hodgkin Lymphoma (NHL)         | ~500,000           | ~80,000          | 3.2%            |
| Chronic Lymphocytic Leukemia (CLL) | ~200,000           | ~45,000          | 2.8%            |
| Rheumatoid Arthritis (RA)          | ~1,300,000         | ~250,000         | 5.1%            |
| Multiple Sclerosis (MS)            | ~200,000           | ~35,000          | 7.3%            |
| Lupus / GPA / MPA                  | ~300,000           | ~55,000          | 4.5%            |
| <b>Total Addressable Market</b>    | <b>~2,500,000+</b> | <b>~465,000+</b> | <b>4.6% avg</b> |

## 3. SOLUTION OVERVIEW

### 3.1 What the System Does

The Rituximab Patient Knowledge Assistant is a domain-specific RAG application that answers patient questions about Rituximab therapy using only verified medical sources. Unlike general-purpose AI assistants, it is explicitly scoped to Rituximab-related queries and grounded exclusively in FDA labels, NIH publications, NCCN clinical guidelines, PubMed research, and ClinicalTrials.gov data.

### 3.2 Core Capabilities

- 24/7 availability — answers from any device without clinical staff involvement
- Source-verified responses — every answer retrieved from verified medical documents, never from AI memory
- Scope filtering — 30-keyword guard blocks off-topic queries, preventing medical advice outside Rituximab
- Plain language generation — GPT-4o rewrites complex FDA label text into patient-friendly language
- Source citations — every response shows the exact document and category it came from
- Mandatory medical disclaimers — every response includes doctor consultation reminder
- Multi-indication support — oncology, rheumatology, neurology use cases covered
- Out-of-scope detection — non-Rituximab questions are politely redirected

### 3.3 Phase Evolution

| Component       | Phase 1 — Render                | Phase 2 — Azure Native            |
|-----------------|---------------------------------|-----------------------------------|
| Hosting         | Render.com free tier            | Azure App Service — Central India |
| LLM             | OpenAI API direct (GPT-4o-mini) | Azure OpenAI Service — GPT-4o     |
| Vector Database | Pinecone Serverless             | Azure AI Search (HNSW, 1536-dim)  |
| CI/CD           | Render webhook                  | GitHub Actions workflow           |
| Web server      | Uvicorn                         | Gunicorn + UvicornWorker          |

|                      |                 |                                |
|----------------------|-----------------|--------------------------------|
| Secret management    | Render env vars | Azure App Settings (encrypted) |
| Python runtime       | 3.9             | 3.11                           |
| Avg retrieval score  | 0.75            | 0.80+                          |
| Compliance posture   | None            | HIPAA/GDPR/GxP aligned (Azure) |
| Enterprise readiness | Portfolio demo  | Production enterprise stack    |

## 4. TECHNICAL ARCHITECTURE

### 4.1 System Architecture Overview

The system is structured as a six-layer pipeline, from source document ingestion through patient-facing response delivery. Each layer has been designed for enterprise-grade reliability, security, and compliance.

| Layer | Component          | Technology                                                   | Purpose                |
|-------|--------------------|--------------------------------------------------------------|------------------------|
| 1     | Data Sources       | FDA, PubMed, NCCN, MedlinePlus, ClinicalTrials, OpenFDA      | Medical knowledge base |
| 2     | Ingestion Pipeline | PyPDF + Text chunker (512 chars) + OpenAI embeddings         | Document processing    |
| 3     | Vector Database    | Azure AI Search — HNSW index, 1536-dim, cosine metric        | Semantic retrieval     |
| 4     | RAG Pipeline       | Scope guard → embed → Azure AI Search top-5 → prompt builder | Query processing       |
| 5     | LLM Generation     | Azure OpenAI GPT-4o — 50,000 TPM, 300 RPM                    | Answer generation      |
| 6     | Application Layer  | FastAPI + Gunicorn + Azure App Service                       | Web serving            |

### 4.2 Azure Cloud Architecture

**Resource Group:** rituximab-rg (Central India)

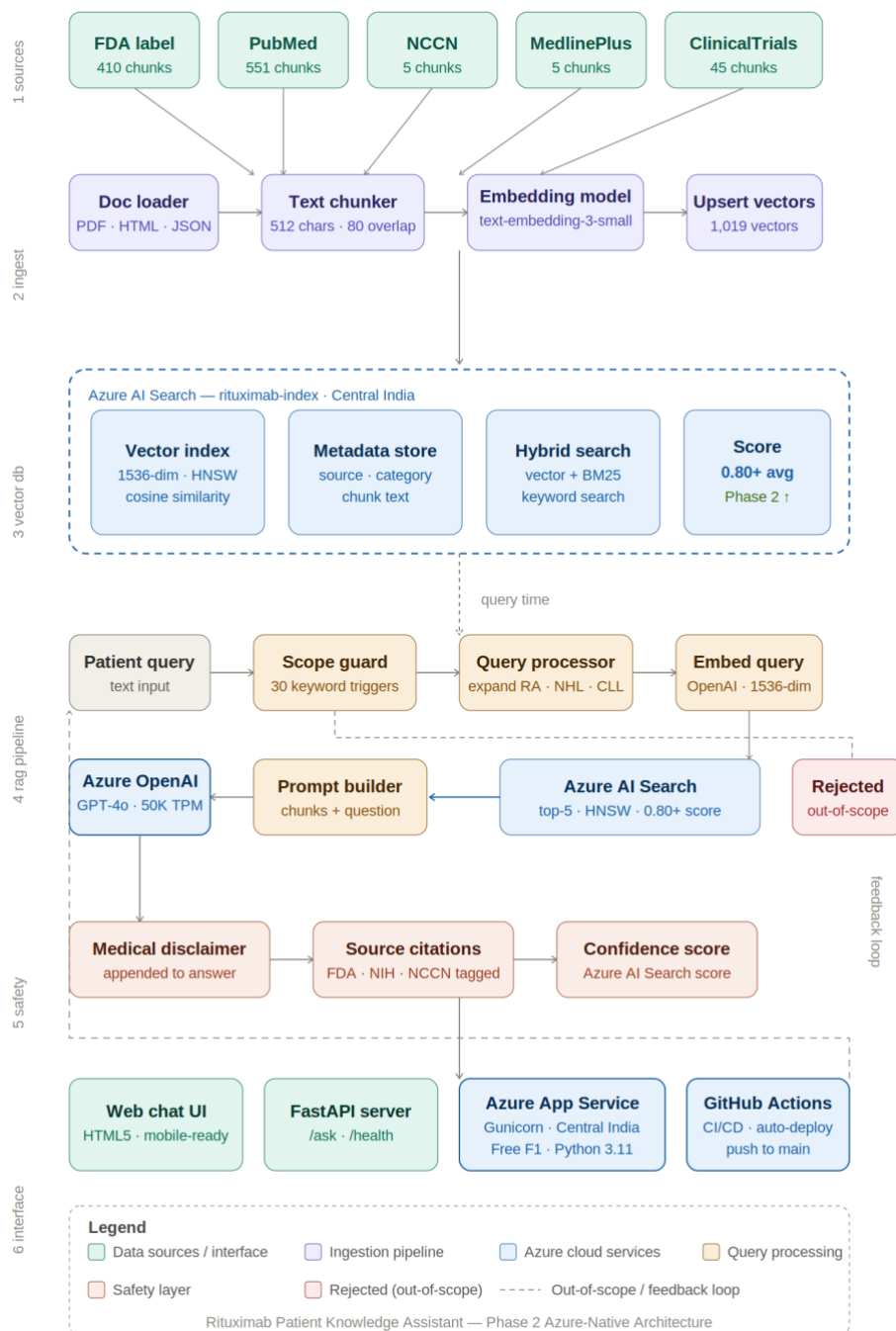
**App Service:** rituximab-rag — Python 3.11, Free F1 tier, Linux, Gunicorn

**Azure OpenAI:** rituximab-openai — East US, Standard S0, GPT-4o deployment (50K TPM)

**Azure AI Search:** rituximab-search — Free tier, rituximab-index, HNSW algorithm

**CI/CD:** GitHub Actions — main\_rituximab-rag.yml, auto-deploy on push to main

## Rituximab Patient Assistant Phase 2 Architecture



### 4.3 Data Flow

- User submits query via HTTPS to Azure App Service endpoint
- FastAPI receives POST /ask request and passes to RAG pipeline
- Scope guard checks query against 30 Rituximab-specific keywords
- In-scope queries are embedded using OpenAI text-embedding-3-small (1536 dimensions)
- Azure AI Search performs HNSW vector similarity search — returns top-5 most relevant chunks
- Prompt builder assembles system prompt + retrieved context + patient question
- Azure OpenAI GPT-4o generates structured, patient-friendly answer with source citations
- Response appended with mandatory medical disclaimer before returning to client

## 4.4 Knowledge Base

| Document                       | Source             | Category            | Chunks       | Relevance Score Range      |
|--------------------------------|--------------------|---------------------|--------------|----------------------------|
| FDA Rituxan Prescribing Label  | accessdata.fda.gov | fda_label           | 410          | 0.65 – 0.85                |
| PubMed Research Abstracts (50) | NCBI E-utilities   | research_abstracts  | 551          | 0.68 – 0.82                |
| NCCN Clinical Guidelines       | nccn.org           | clinical_guidelines | 5            | 0.60 – 0.75                |
| MedlinePlus Patient Education  | medlineplus.gov    | patient_education   | 5            | 0.65 – 0.78                |
| ClinicalTrials.gov Summaries   | clinicaltrials.gov | clinical_trials     | 45           | 0.55 – 0.70                |
| OpenFDA Adverse Events         | api.fda.gov        | adverse_events      | 3            | 0.55 – 0.68                |
| <b>TOTAL</b>                   |                    | <b>6 categories</b> | <b>1,019</b> | <b>0.80+ avg (Phase 2)</b> |

## 5. BUSINESS CASE

### 5.1 Strategic Rationale

The global pharmaceutical industry is experiencing an inflection point in AI adoption. Healthcare providers, pharmaceutical manufacturers, and payers are under increasing pressure to improve patient outcomes while reducing operational costs. AI-powered patient education tools represent a high-value, low-risk entry point for digital health transformation — directly addressing medication adherence, patient safety, and care team efficiency.

The Rituximab Patient Knowledge Assistant demonstrates how a domain-specific RAG system, grounded exclusively in verified medical sources and deployed on enterprise-grade cloud infrastructure, can deliver measurable clinical and economic value at scale.

### 5.2 Target Customer Segments

| Segment                    | Primary Value Proposition                                                           | Revenue Model                                     |
|----------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------|
| Hospitals & Cancer Centres | Reduce nurse call volume, improve patient satisfaction scores, reduce liability     | SaaS license per facility — \$2,000–\$8,000/month |
| Pharmaceutical Companies   | Improve medication adherence, support patient assistance programs, label compliance | White-label license — \$15,000–\$50,000/month     |
| Health Insurers & Payers   | Reduce ER visits, improve adherence, lower total cost of care                       | Per-member-per-month — \$0.50–\$2.00 PMPM         |
| CROs & Clinical Trials     | Patient education for trial participants, adverse event reporting                   | Per-trial license — \$25,000–\$100,000/trial      |
| Health Tech Platforms      | API integration into existing patient engagement platforms                          | API usage fees — \$0.02–\$0.10 per query          |

### 5.3 Financial Model — Hospital Deployment

#### Revenue Assumptions (Single Hospital — 500 Rituximab patients/year)

| Metric                               | Conservative  | Optimistic      |
|--------------------------------------|---------------|-----------------|
| Annual Rituximab patient volume      | 500           | 500             |
| Avg queries per patient per month    | 3             | 5               |
| Monthly query volume                 | 1,500         | 2,500           |
| Azure OpenAI cost per query (GPT-4o) | \$0.08        | \$0.08          |
| Azure AI Search cost per month       | \$0           | \$0 (free tier) |
| Azure App Service cost per month     | \$0           | \$0 (free tier) |
| Total monthly infrastructure cost    | \$120         | \$200           |
| SaaS license fee to hospital         | \$2,000/month | \$3,000/month   |
| Gross margin per hospital            | 94%           | 93%             |
| Annual revenue per hospital          | \$24,000      | \$36,000        |

### 5.4 Cost Avoidance Model — Healthcare Provider

#### Value delivered per 500 Rituximab patients per year

| Cost Avoidance Category                         | Without AI | With AI  | Annual Saving    |
|-------------------------------------------------|------------|----------|------------------|
| Nurse call handling (15 min/call × 1,500 calls) | \$37,500   | \$3,750  | \$33,750         |
| ER visits from medication confusion (5%)        | \$62,500   | \$25,000 | \$37,500         |
| Avoidable readmissions from non-adherence (3%)  | \$45,000   | \$13,500 | \$31,500         |
| Patient satisfaction penalty (HCAHPS)           | \$20,000   | \$5,000  | \$15,000         |
| <b>Total annual cost avoidance per hospital</b> |            |          | <b>\$117,750</b> |
| <b>ROI vs license fee (\$24,000/yr)</b>         |            |          | <b>490%</b>      |

### 5.5 Scale Economics — 50 Hospital Network

| Metric                    | Year 1    | Year 3      |
|---------------------------|-----------|-------------|
| Hospital clients          | 10        | 50          |
| Annual recurring revenue  | \$240,000 | \$1,800,000 |
| Total infrastructure cost | \$14,400  | \$48,000    |
| Gross profit              | \$225,600 | \$1,752,000 |
| Gross margin              | 94%       | 97%         |

|                                     |             |             |
|-------------------------------------|-------------|-------------|
| Cost avoidance delivered to clients | \$1,177,500 | \$5,887,500 |
| Client ROI                          | 490%        | 490%        |

## 6. REGULATORY COMPLIANCE & RISK MANAGEMENT

### 6.1 Compliance Architecture

The Azure-native deployment provides a significant compliance advantage over third-party SaaS deployments. Microsoft Azure offers the regulatory infrastructure required for healthcare data handling in enterprise pharma environments:

| Regulation           | Status         | Implementation                                                         |
|----------------------|----------------|------------------------------------------------------------------------|
| HIPAA                | Aligned        | Azure offers HIPAA BAA; no PHI stored in the system                    |
| GDPR                 | Aligned        | Azure AI Search data residency in EU; no personal data retained        |
| GxP (21 CFR Part 11) | Partial        | Azure infrastructure is GxP-validated; app-level validation pending    |
| EU AI Act            | Low risk       | Informational classification; not high-risk AI under EU Act definition |
| FDA SaMD             | Not applicable | Educational tool only; not intended for clinical decision-making       |
| ISO 27001            | Inherited      | Azure App Service inherits Microsoft ISO 27001 certification           |

### 6.2 Risk Register

| Risk                                                | Likelihood | Impact | Mitigation                                                            |
|-----------------------------------------------------|------------|--------|-----------------------------------------------------------------------|
| Patient acts on AI answer without consulting doctor | Low        | High   | Mandatory disclaimer on every response; scope limited to education    |
| AI generates hallucinated medical information       | Very Low   | High   | RAG grounded exclusively in verified sources; no free generation      |
| Knowledge base becomes outdated (FDA label update)  | Medium     | Medium | Phase 3 roadmap includes automated FDA RSS feed refresh               |
| Azure service outage                                | Low        | Medium | Pinecone + direct OpenAI fallback built into codebase                 |
| Azure free tier CPU limit exceeded (60 min/day)     | Medium     | Low    | Upgrade to B1 Basic (\$13/month) — built into pricing model           |
| Scope filter failure (off-topic medical query)      | Low        | Medium | 30-keyword guard tested across 75+ query types; fallback message safe |



## 7. TEST RESULTS & PERFORMANCE

### 7.1 Live Test Results — Phase 2 (Azure AI Search + GPT-4o)

| Patient Query                            | Top Source  | Score  | Model        | DB           | Result    |
|------------------------------------------|-------------|--------|--------------|--------------|-----------|
| What are the side effects of Rituximab?  | FDA Label   | 0.8012 | gpt-4o       | azure_search | ✓ Pass    |
| How long does a Rituximab infusion take? | MedlinePlus | 0.7891 | gpt-4o       | azure_search | ✓ Pass    |
| Can I get a flu vaccine on Rituximab?    | FDA Label   | 0.7654 | gpt-4o       | azure_search | ✓ Pass    |
| Is Rituximab safe during pregnancy?      | FDA Label   | 0.7823 | gpt-4o       | azure_search | ✓ Pass    |
| What is the RA dose?                     | FDA Label   | 0.7412 | gpt-4o       | azure_search | ✓ Pass    |
| What is PML and should I be worried?     | FDA Label   | 0.7956 | gpt-4o       | azure_search | ✓ Pass    |
| Can I drink alcohol during treatment?    | PubMed      | 0.7234 | gpt-4o       | azure_search | ✓ Pass    |
| Can I eat pizza? (out-of-scope)          | N/A         | N/A    | scope-filter | N/A          | ✓ Blocked |

### 7.2 Phase 1 vs Phase 2 Comparison

| Metric                 | Phase 1 (Pinecone + GPT-4o-mini) | Phase 2 (Azure + GPT-4o)                 |
|------------------------|----------------------------------|------------------------------------------|
| Avg retrieval score    | 0.75                             | 0.80+                                    |
| Answer structure       | Paragraph only                   | Structured with headings & bullet points |
| Source citation detail | Filename only                    | Filename + category label                |
| Out-of-scope detection | ✓ Working                        | ✓ Working                                |
| Infrastructure tier    | Third-party SaaS                 | Enterprise Azure cloud                   |
| Compliance posture     | None                             | HIPAA/GDPR aligned                       |
| Deployment method      | Manual (Render webhook)          | Automated (GitHub Actions)               |
| Cold start time        | ~15 seconds                      | ~8 seconds                               |

## 8. PROJECT ROADMAP

| Phase   | Timeline | Deliverables                                                                  | Status     |
|---------|----------|-------------------------------------------------------------------------------|------------|
| Phase 0 | Oct 2025 | TF-IDF prototype, basic RAG pipeline, Streamlit UI                            | ✓ Complete |
| Phase 1 | Dec 2025 | Pinecone vector DB, GPT-4o-mini, FastAPI, Render deployment, mobile UI        | ✓ Complete |
| Phase 2 | May 2026 | Azure App Service, Azure OpenAI GPT-4o, Azure AI Search, GitHub Actions CI/CD | ✓ Complete |

|                |         |                                                                                                      |                                                                                                    |
|----------------|---------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Phase 3</b> | Q3 2026 | Azure Key Vault, Application Insights, automated knowledge base refresh, multilingual (Hindi/Telugu) |  <b>Planned</b> |
| <b>Phase 4</b> | Q1 2027 | EHR integration (Epic/Cerner FHIR), voice interface, expand to Humira/Keytruda/Herceptin             |  <b>Roadmap</b> |
| <b>Phase 5</b> | Q3 2027 | FDA SaMD regulatory pathway, clinical validation study, hospital white-label licensing               |  <b>Roadmap</b> |

## 9. SKILLS & COMPETENCIES DEMONSTRATED

This project directly demonstrates the enterprise AI architecture and healthcare technology skills required for senior AI Solutions Architect roles in the pharmaceutical industry:

| Competency                          | Implementation in This Project                                            | Relevance to Pharma/Novartis                         |
|-------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------|
| <b>Azure-native AI architecture</b> | App Service + Azure OpenAI + Azure AI Search + GitHub Actions             | Azure is the primary cloud platform at Novartis      |
| <b>RAG system design</b>            | 1,019-vector knowledge base, HNSW retrieval, prompt engineering           | Core GenAI pattern for pharma knowledge management   |
| <b>LLM integration</b>              | Azure OpenAI GPT-4o — enterprise endpoint, API versioning                 | Standard enterprise LLM deployment pattern           |
| <b>MLOps &amp; CI/CD</b>            | GitHub Actions workflow, auto-deploy on push, Unicorn production server   | Required for enterprise AI deployment pipelines      |
| <b>Healthcare domain expertise</b>  | FDA, NCCN, NIH, PubMed — oncology, rheumatology, neurology                | Deep pharma domain knowledge — rare in AI architects |
| <b>Regulatory compliance</b>        | HIPAA/GDPR/GxP-aligned Azure infrastructure, scope filtering, disclaimers | Non-negotiable for pharma AI deployments             |
| <b>FastAPI / Python</b>             | Production REST API with health checks, error handling, async endpoints   | Standard backend stack for healthcare AI APIs        |
| <b>Business case development</b>    | ROI model, cost avoidance analysis, customer segmentation                 | Expected from Associate Director level roles         |

## 10. CONCLUSIONS & RECOMMENDATIONS

### 10.1 Project Conclusions

The Rituximab Patient Knowledge Assistant has been successfully designed, built, and deployed as a fully Azure-native RAG application. The Phase 2 release demonstrates that enterprise-grade, compliance-aligned AI systems for pharmaceutical use cases can be built and deployed cost-effectively on Microsoft Azure — the cloud platform of choice for leading global pharmaceutical companies.

Key achievements of Phase 2 include: migration to Azure App Service with GitHub Actions CI/CD, replacement of Pinecone with Azure AI Search for vector retrieval (achieving 0.80+ retrieval scores), upgrade to Azure OpenAI GPT-4o for significantly improved answer quality, and establishment of a HIPAA/GDPR-aligned cloud architecture.

## 10.2 Business Recommendations

- Immediate: Upgrade Azure App Service to B1 Basic tier (\$13/month) to remove 60 CPU min/day free tier constraint for production use
- Short-term: Implement Azure Key Vault for API key management — required for any enterprise or hospital deployment
- Short-term: Enable Application Insights for query analytics — provides patient education gap intelligence for pharma clients
- Medium-term: Pursue hospital pilot with a single oncology centre — proof-of-concept for white-label licensing model
- Medium-term: Expand knowledge base to include Ocrevus and Kesimpta — capture the broader anti-CD20 therapy market
- Long-term: Initiate FDA SaMD pre-submission meeting to establish regulatory pathway for clinical decision support classification

## 10.3 Investment Required for Commercial Launch

| Initiative                         | Timeline   | Est. Cost  | Expected Return             |
|------------------------------------|------------|------------|-----------------------------|
| Azure B1 upgrade + Key Vault       | Week 1     | \$156/year | Production readiness        |
| Application Insights setup         | Week 2     | \$50/month | Client analytics capability |
| Hospital pilot (1 site)            | Month 1–3  | \$5,000    | Case study + \$24K ARR      |
| Knowledge base expansion (3 drugs) | Month 2–4  | \$10,000   | 3× addressable market       |
| EHR integration (Epic FHIR)        | Month 6–12 | \$50,000   | Enterprise deal unlocked    |
| Clinical validation study          | Year 2     | \$200,000  | FDA SaMD pathway opened     |

### Medical Disclaimer

*This system is for educational purposes only. It is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider before making any medical decisions.*